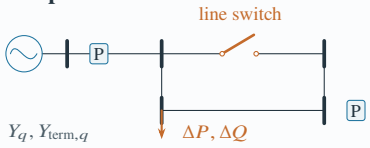
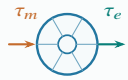
PHYSICAL ATTRIBUTION

Graph structure → dynamic histories → Bayesian attribution

Graph: electrical interconnection



Hamiltonian: rotor storage



$K = \sum_g S_g H_g \omega_g^2$

Torque-driven rotor subsystem

$\dot{K} = \sum_g S_g \omega_g [\tau_{m,g} - \tau_{e,g} - D_g(\omega_g - 1)]$

Sparse PMU observations



$|V|, \angle V, |I|, \angle I, f, \dot{f}$

8 PMUs; 47 coordinates; 30 reports/s

Oriented terminals + causal instrument model

Implemented core

Same physical state

new report r_k

1 Physical operators

$\dot{x} = f_q(x, z, u)$

$0 = g_q(x, z, u)$



branch stamps → reduced maps

$T_o, J_o, g_o, u_o, j_o, v_o$

Prefix-fit context

Fixed network alternatives

2 History bank

No new event

Recovery

New load or line

$(m_h, d_h, \mu_h, v_h, \pi_h)$

State–severity dependence

3 Bayesian update



$e_h = r_k - \hat{r}_{h,k}$

Joint covariance

$L_{h,k} = \int p(r_k \mid h, a)\, p_h(a)\, da$

$\pi_{h,k} \propto L_{h,k}\, \pi_{h,k}^{\text{prior}}$

Normalize; prune; retain state

4 Publish separately



M_c: active Γ_c: cause

CAL threshold + confirmation

Type / source / restoration

No reset at physical OFF

Shared covariance; prefix-fitted noise

Next report: state and history weights

REFINEMENT DEVELOPMENT

Mechanism diagnostics; integration into the history bank remains pending

Preserve recovery uncertainty



Same mean

Different predictive spread

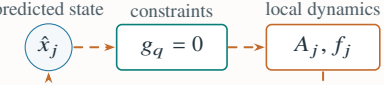
$P_{\text{mix}} = \sum_h \pi_h [P_h + (\mu_h - \bar{\mu})(\mu_h - \bar{\mu})^\top]$

Adapt noise without hiding events



Score first; update after. Keep center fixed.

Refresh physical predictions



Retain state; recompute local derivatives

CYBER LAYER

External methods + telemetry → Quality / availability interface → Separate alerts and validation

Drawings are schematic. Cyber scores are not extra physical evidence. Dashed modules are not enabled in the reported history-bank runs.